CELL REPRODUCTION MASTER STUDY GUIDE

A Clear, Detailed, and Visual Guide to How Cells Grow, Duplicate, and Divide

1. Foundations of Cell Reproduction & The Centromere Rule

Cell division is how living things grow, repair damaged body parts, and pass life to the next generation. In regular human body cells (called **somatic cells**), we have **46 chromosomes** arranged in 23 pairs—one set inherited from mom and one set from dad (this double set is called **diploid**, written as *2n = 46*).

The Centromere Rule (How to Count Chromosomes Easily):
Students often get confused about chromosome math during cell division. Here is the golden rule: **Always count the centromeres!**
• *Before DNA copies:* A cell has 46 single-stranded chromosomes (46 centromeres, 46 chromatids).
• *After Interphase (DNA duplication):* The cell makes an exact copy of every chromosome. The two identical copies stay attached at a central button called the **centromere**. Because they are connected at 1 centromere, we still count it as **46 chromosomes**, even though there are now **92 sister chromatids**!
• *When Sister Chromatids Separate:* As soon as the twin chromatids pull apart, each individual strand becomes its own independent chromosome with its own centromere.

Stage / Phase	Human Cell State	Centromere Count	Total Chromatids
Before Interphase	Single-stranded (2n)	46 Chromosomes	46 Chromatids
After Interphase (Replicated)	Double-stranded 'X' shape	46 Chromosomes	92 Sister Chromatids
After Mitosis Split	2 Identical Daughter Cells	46 Chromosomes / cell	46 Chromatids / cell
After Meiosis I Split	2 Haploid Cells (n)	23 Chromosomes / cell	46 Sister Chromatids
After Meiosis II Split	4 Unique Gametes (n)	23 Chromosomes / cell	23 Chromatids / cell

2. Bacterial Binary Fission & Asexual Reproduction

Binary Fission (Bacterial Quick-Cloning): Single-celled prokaryotes like bacteria reproduce without a nucleus or spindle fibers through a swift 5-step process:
1. **DNA Replication:** Begins at the *origin of replication*, copying the circular chromosome in both directions.
2. **Cell Elongation:** The cell stretches longer while a specialized division protein called **FtsZ** migrates toward the midpoint.
3. **Chromosome Segregation & Z-Ring:** The two duplicate circular chromosomes pull apart to opposite ends, and FtsZ forms a tight **Z-ring** at the middle.
4. **Septum Wall Construction:** A new cell wall called a **septum** grows inward along the Z-ring.
5. **Cell Splitting:** The membrane pinches closed, splitting the parent into 2 completely identical daughter bacteria clones.

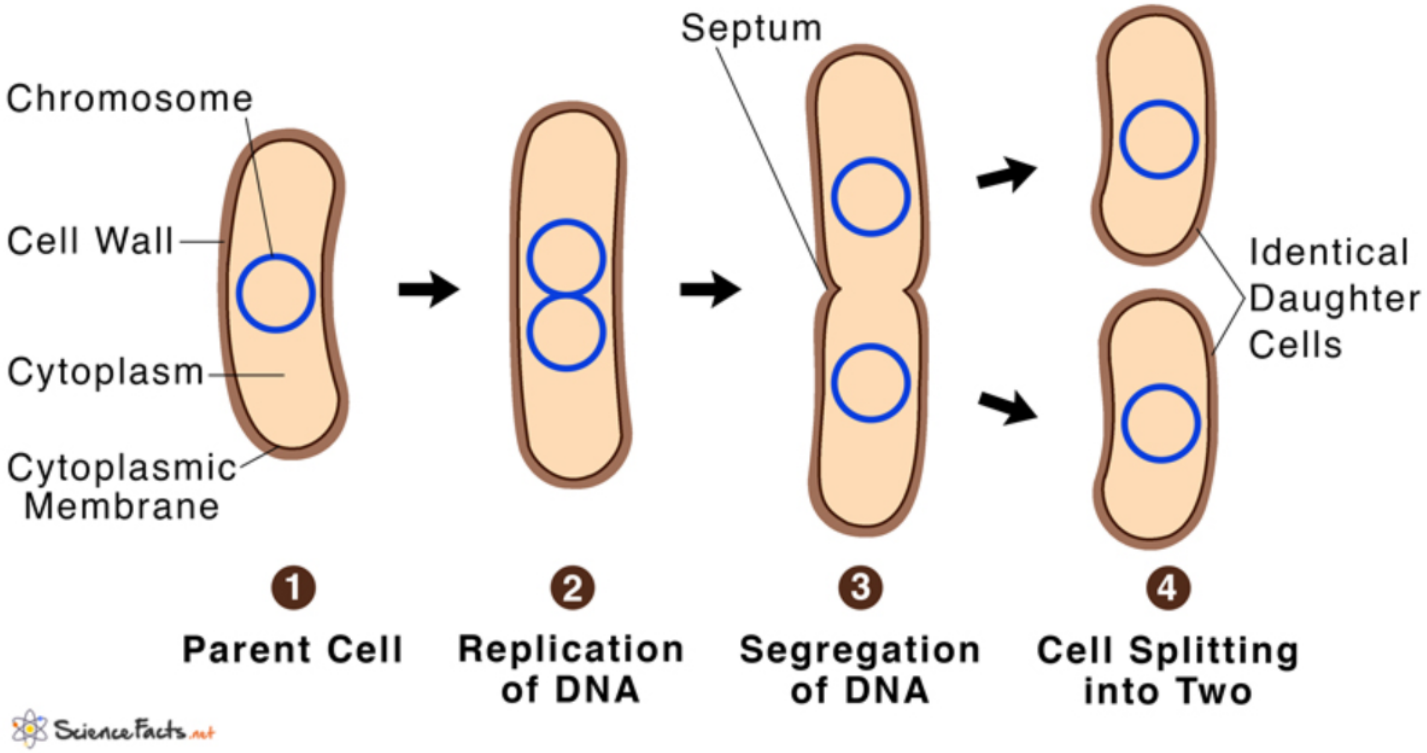


Figure 1: Bacterial Binary Fission Process — DNA Replication, Segregation, Septum Formation, and Division (Source: Image1.png)

Other Asexual Mechanisms (1 Parent = Clones):

• **Budding:** A tiny bump (bud) grows directly on the parent's body through cell division and breaks off when mature (used by yeast, *Hydra*, jellyfish, and stalked bacteria like *Hyphomicrobium*).

• **Vegetative Propagation:** Plants growing complete new identical plants from stems, roots, or leaf buds (e.g., potato eyes growing new stalks).

• **Fragmentation:** An organism breaks into distinct pieces, and each piece regenerates missing parts to form a complete clone (e.g., California blackworms).

3. Mitosis: Your Body's Exact Copy Machine

Purpose: Mitosis takes 1 diploid body cell (2n = 46) and makes **2 genetically identical daughter cells**. It is used for growth, replacing worn-out cells, and healing cuts or tissue damage.

The Sequential PMAT Stages:

• **Interphase (Preparation):** The cell grows, performs normal metabolic functions, and duplicates its DNA so chromosomes become double-stranded X-shapes.

• **Prophase (Preparation):** Chromosomes pack tight and thicken (condense) so they don't tangle. The nuclear envelope dissolves, and barrel-like centrioles move to opposite ends, forming spindle fibers.

• **Metaphase (Middle):** Microtubule spindle fibers attach to the centromeres, pulling chromosomes into a neat **single-file line** right down the center (equator) of the cell.

• **Anaphase (Away):** Spindle fibers shorten, snapping the centromeres and pulling sister chromatids **away** to opposite sides of the cell. Each separated chromatid is now an independent chromosome.

• **Telophase & Cytokinesis (Two Nuclei & Cell Split):** Two new nuclear envelopes rebuild around the separated chromosome groups. An actin/myosin contractile ring pinches the cell membrane inward (forming a *cleavage furrow*) to split the cytoplasm completely in two.

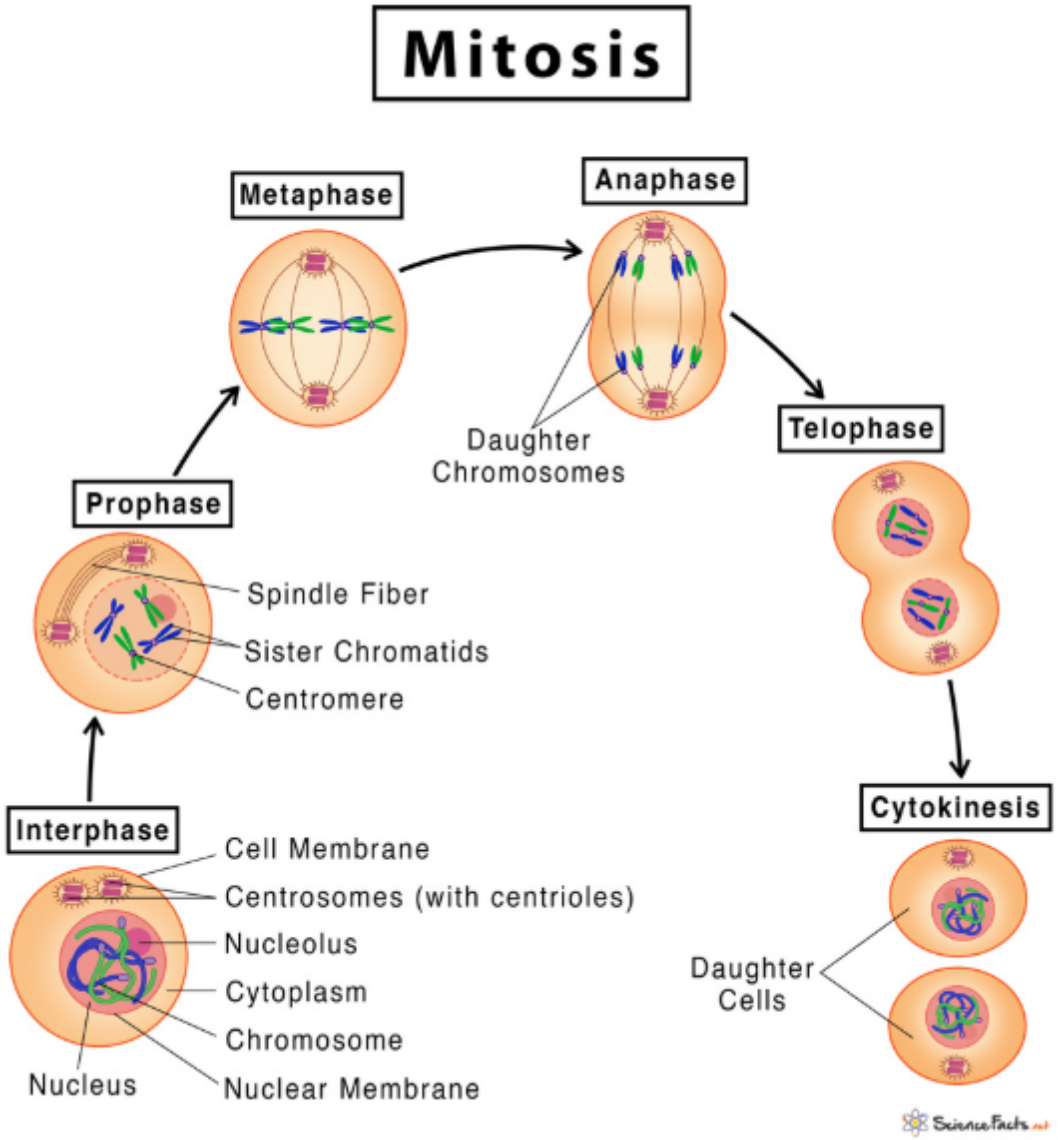


Figure 2: Complete Mitotic Cell Cycle — Interphase through PMAT and Cytokinesis (Source: Image2.png)

4. Meiosis: Nature's Genetic Shuffle & Cut Machine

Purpose: Meiosis makes specialized reproductive sex cells (**gametes**: sperm and eggs). It reduces the chromosome count by half—turning 1 diploid cell (46 chromosomes) into **4 genetically unique haploid cells** (23 single chromosomes each). This ensures that when sperm (23) meets egg (23) during fertilization, the resulting baby has the correct 46 chromosomes.

Why Meiosis Divides Twice (PMAT I & PMAT II):

• **Meiosis I (Halving the Pairs):**

- *Prophase I:* Homologous chromosome pairs match up side-by-side. They touch and swap genetic pieces in a critical process called **Crossing Over**! This recombines mom's and dad's genes, creating unique trait combinations for every sibling.
- *Metaphase I:* Homologous pairs line up down the middle in **double-file pairs**. Which side mom's or dad's chromosome faces is completely random (*Independent Assortment*).
- *Anaphase I:* Homologous pairs get pulled away to opposite poles, but sister chromatids stay attached.
- *Telophase I & Cytokinesis I:* Splits the cell into 2 haploid cells (each with 23 duplicated chromosomes).

• **Meiosis II (Splitting the Sister Chromatids):**

- Operates like regular mitosis! Chromosomes line up **single-file** in Metaphase II, and sister chromatids pull apart in

Anaphase II. The final result is **4 genetically unique haploid gametes**.

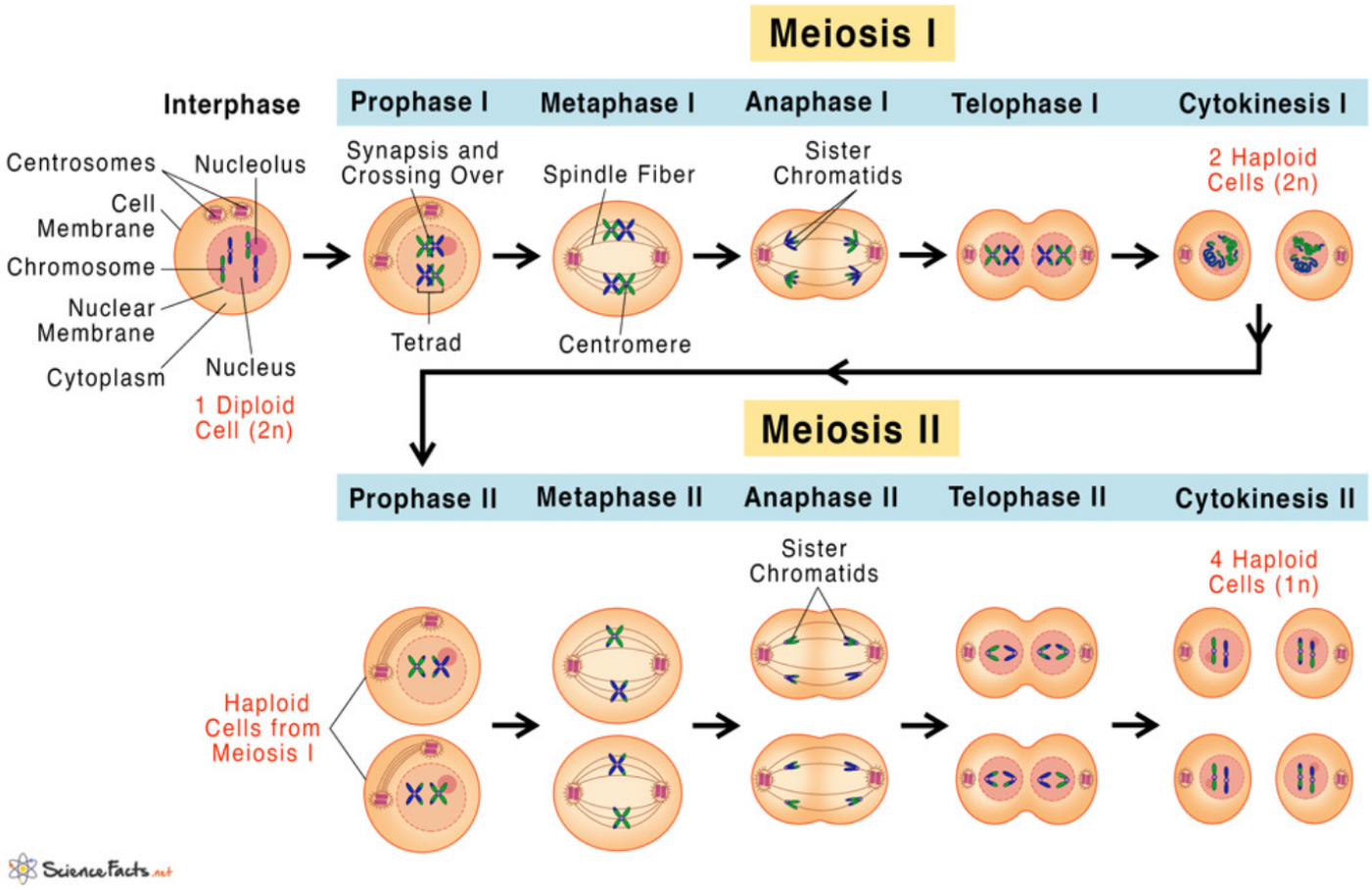


Figure 3: Meiosis I and II Double-Division Stages — Crossing Over, Homologous Separation, and Chromatid Splitting (Source: Image3.png)

5. Detailed Side-by-Side Comparison: Mitosis vs. Meiosis

Comparing Mitosis and Meiosis side-by-side highlights their fundamental biological differences:

1. **Biological Objective:** Mitosis creates identical clones for body growth and repair. Meiosis creates varied sex cells (gametes) for sexual reproduction.
2. **Number of Divisions:** Mitosis undergoes **1 division** round (1 PMAT). Meiosis undergoes **2 division** rounds (PMAT I and PMAT II).
3. **Metaphase Line-Up Formation:** In Mitosis Metaphase, chromosomes line up **single-file** down the middle. In Meiosis Metaphase I, chromosomes line up in **double-file homologous pairs**.
4. **Genetic Variation Mechanisms:** Mitosis has *zero genetic mixing*—sister chromatids are exact duplicates. Meiosis creates massive genetic variation through **Crossing Over** in Prophase I and **Independent Assortment** in Metaphase I.
5. **Chromosome & Cell Outcome:** Mitosis yields **2 Diploid (2n = 46)** identical body cells. Meiosis yields **4 Haploid (n = 23)** unique sex cells.

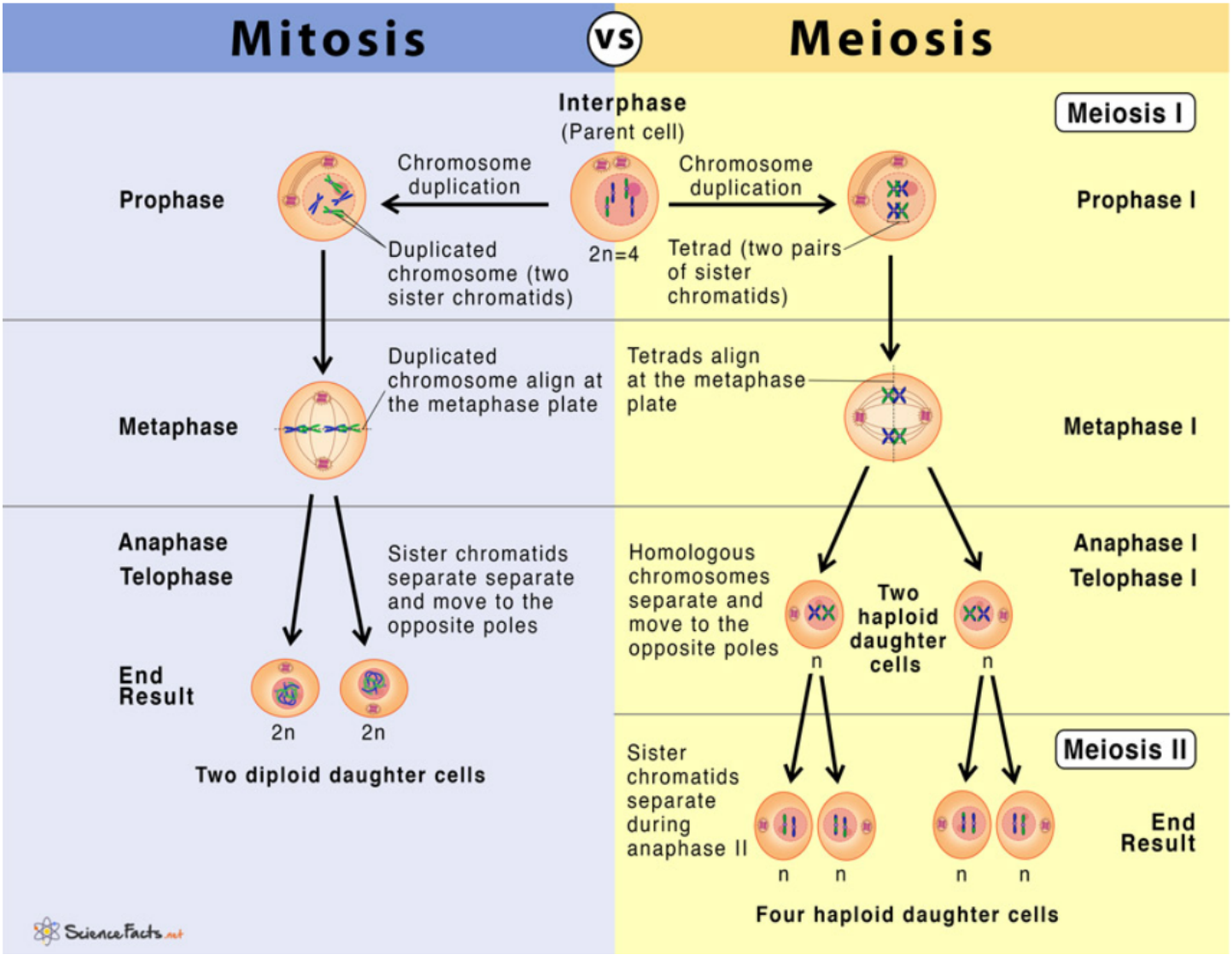


Figure 4: Side-by-Side Architectural Comparison of Mitosis vs. Meiosis (Source: Image4.png)

Feature	Binary Fission	Mitosis	Meiosis
Cell Type	Prokaryotes (Bacteria)	Eukaryotes (Somatic Body Cells)	Eukaryotes (Germ Cells / Gonads)
Primary Purpose	Asexual Reproduction	Growth, Repair, & Healing	Making Sex Cells (Sperm & Eggs)
Division Rounds	1 Fast Division	1 Division (PMAT)	2 Divisions (PMAT I & PMAT II)
Daughter Cells	2 Identical Clones	2 Identical Clones	4 Genetically Unique Gametes
Ploidy / Count	1 Circular Chromosome	Diploid (2n = 46)	Haploid (n = 23)
Genetic Variety	None (Direct Clones)	None (Exact Copies)	High (Crossing Over & Assortment)

6. Human Gametogenesis: Spermatogenesis vs. Oogenesis

Spermatogenesis (Sperm Production in Males):

• Starts at puberty inside the **seminiferous tubules** of the testes and continues throughout life.

• Diploid precursor cells (*spermatogonia*) undergo meiosis to produce 4 equal, immature haploid *spermatids*.

• Through *spermiogenesis*, spermatids grow a swimming flagellum, a energy-rich middle piece, and an **acrosome cap** containing enzymes to penetrate the egg, yielding **4 active, motile spermatozoa** (each with 23 chromosomes: 22 autosomes

+ X or Y).

Oogenesis (Egg Production in Females):

• Begins inside the ovaries of a female fetus before birth! All primary oocytes pause at **Prophase I** until puberty.

• Starting at puberty, 1 oocyte resumes Meiosis I each month, dividing cytoplasm unevenly to make **1 large secondary oocyte** (packed with nutrients) and 1 tiny non-functional **first polar body**.

• The secondary oocyte pauses again at **Metaphase II** and is released during ovulation. Meiosis II finishes *only if a sperm fertilizes the egg*, yielding **1 mature ovum** (22 autosomes + X) and a second polar body that dissolves.

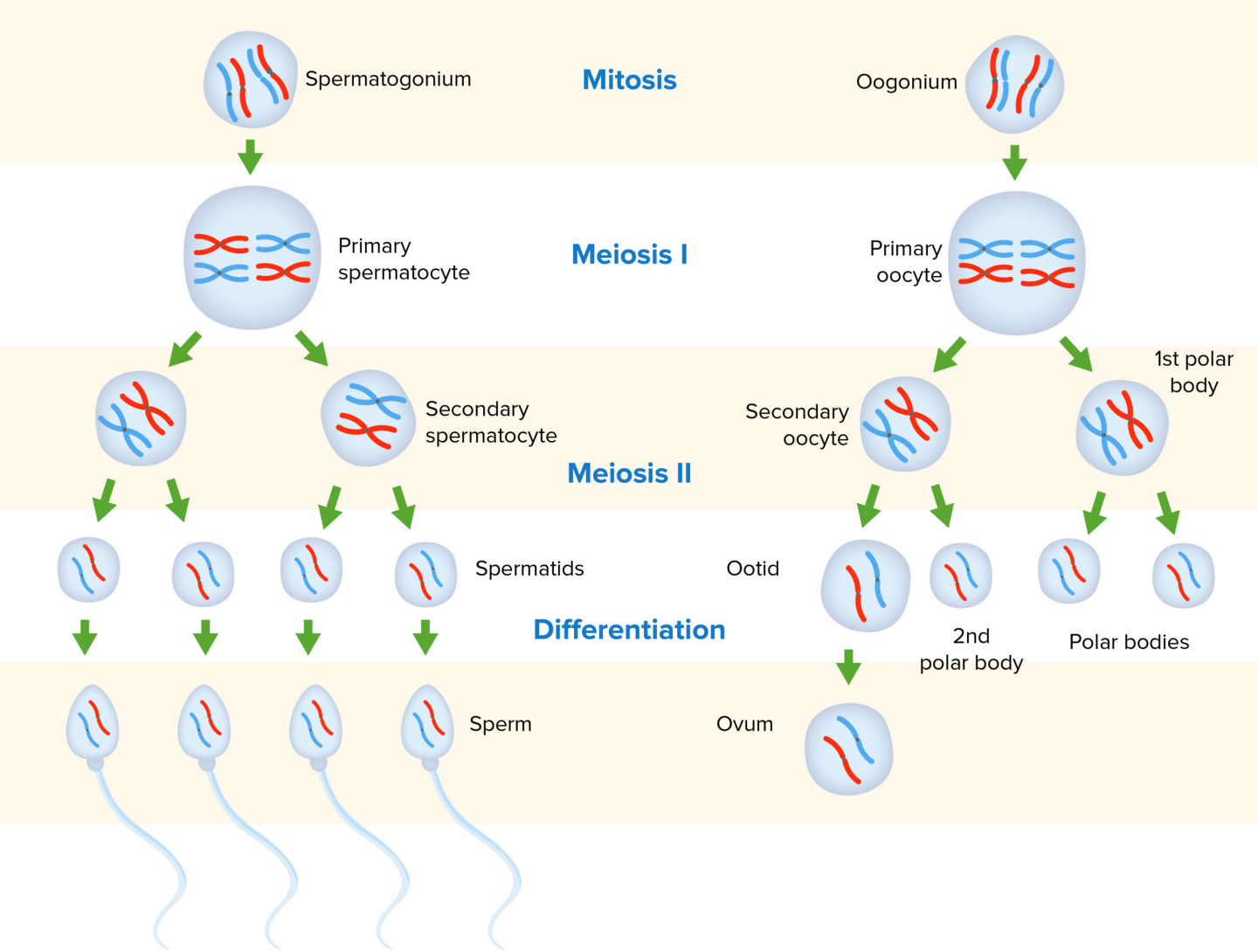


Figure 5: Human Gametogenesis Pathways — Spermatogenesis vs. Oogenesis Developmental Comparison (Source: Image5.jpg)

HUMAN REPRODUCTION SYSTEM & MENSTRUATION GUIDE

A Comprehensive Reference Manual on Anatomy, Hormonal Regulation, Clinical Conditions, and Embryogenesis

1. Male Reproductive System Anatomy & Physiology

Human reproduction relies on specialized male and female anatomical structures designed to produce gametes (egg and sperm cells), facilitate fertilization, and nurture developing offspring. The male reproductive system is specialized for continuous sperm production (spermatogenesis), maturation, fluid nourishment, and efficient delivery during copulation.

A. Anterior (Front View) Anatomical Profile

When viewed from an anterior (front) anatomical perspective, the male reproductive system demonstrates a symmetrical, interconnected network of gonadal, tubular, and glandular components. The primary male gonads—the testicles—are situated externally, while internal transport ducts and accessory glands converge near the urinary bladder base to form semen.

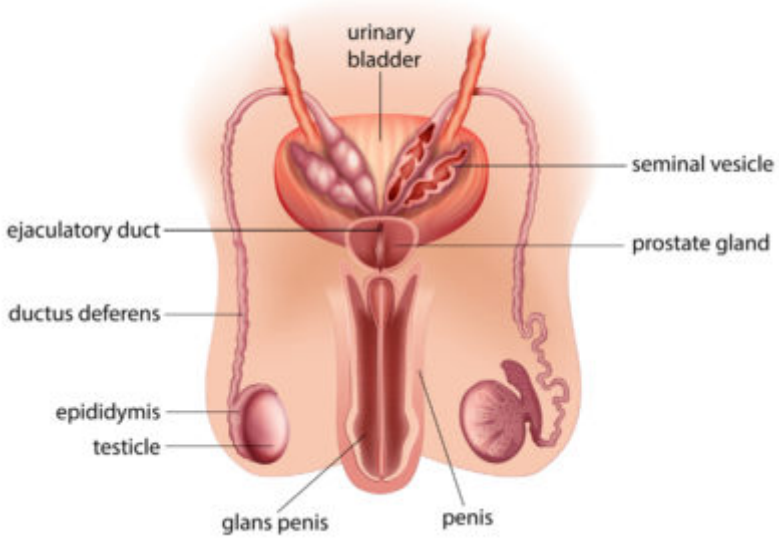


Figure 1A: Male Reproductive System — Anterior (Front View) Anatomical Diagram (Image-1).

Anatomical Structure	Physiological Function & Characteristics
Urinary bladder	Muscular sac situated superior to the prostate; temporarily stores urine before excretion.
Seminal vesicle	Paired glands posterior to the bladder; secrete an alkaline, fructose-rich fluid making up ~60% of semen.
Ejaculatory duct	Short junction duct formed by vas deferens and seminal vesicle; passes through the prostate to enter the urethra.
Prostate gland	Encircles the urethra immediately below the bladder; secretes milky, alkaline fluid to enhance sperm motility.
Ductus deferens (Vas deferens)	Muscular transport tube carrying mature sperm upward from the epididymis into the pelvic cavity.
Epididymis	Coiled crescent-shaped storage structure atop each testicle where immature sperm gain motility over 2–3 weeks.
Testicle (Testis)	Primary male gonad located in the scrotum; secretes testosterone and manufactures sperm cells.
Penis and glans penis	External copulatory organ and its sensitive bulbous tip (glans) facilitating semen delivery and urination.

B. Sagittal (Side View) Profile & Duct Pathways

The sagittal (side view) anatomical perspective illustrates the spatial relationship between the male reproductive duct system, pelvic bone landmarks, and the digestive tract. Spermatogenesis takes place inside microscopic, highly coiled **seminiferous tubules** within the testes. Emerging sperm enter the **epididymis** for structural maturation and storage.

During sexual arousal, rhythmic peristaltic contractions propel sperm through the **vas deferens**, looping over the urinary bladder toward the ejaculatory ducts. Here, secretions from the **seminal vesicles** (providing energy-rich fructose) and the **prostate gland** (providing neutralizing fluid) combine with sperm cells. Finally, the **bulbourethral (Cowper's) glands** contribute a pre-ejaculatively secreted clear fluid that lubricates the urethra and neutralizes residual urinary acidity prior to ejaculation.

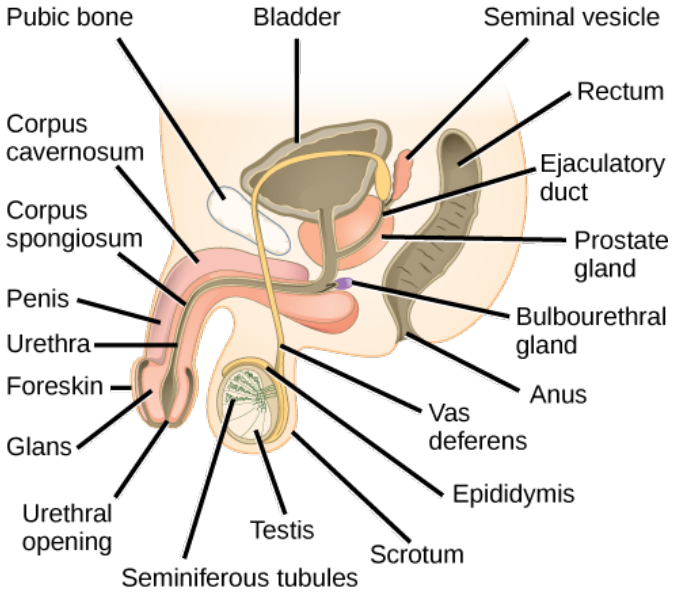


Figure 1B: Male Reproductive System — Sagittal (Side View) Anatomical Profile (Figure_43_03_01.png).

Anatomical Structure	Physiological Function & Structural Context
Testis	Male gonad responsible for producing sperm cells and secreting testosterone.
Seminiferous tubules	Highly coiled microscopic tubules within the testis where spermatogenesis occurs.
Epididymis	Coiled duct atop the testis where sperm mature, acquire motility, and are stored.
Scrotum	External skin pouch enclosing the testes; regulates testicular temperature ~2–3°C below core body heat.
Vas deferens	Muscular tube propelling mature sperm from the epididymis into the pelvic cavity.
Seminal vesicle	Gland secreting alkaline fluid enriched with fructose, prostaglandins, and clotting proteins.
Ejaculatory duct	Short duct where the vas deferens merges with the seminal vesicle duct inside the prostate.
Prostate gland	Chestnut-sized gland surrounding the prostatic urethra; secretes alkaline, zinc-rich fluid.
Bulbourethral gland	Pea-sized glands secreting clear mucus to lubricate the urethra and neutralize acid.
Bladder	Temporary storage reservoir for urine located superior to the prostate gland.
Urethra	Dual-function fibromuscular tube carrying either urine or semen through the penis.
Urethral opening	Terminal external meatus at the tip of the glans penis for fluid exit.
Penis	External copulatory organ containing erectile tissue (corpus cavernosum and spongiosum).
Pubic bone, rectum, anus	Surrounding skeletal and gastrointestinal landmarks providing anatomical orientation.

2. Female Reproductive System Anatomy & Physiology

The female reproductive system is specialized for producing female gametes (ova), synthesizing steroid sex hormones (estrogen and progesterone), supporting fertilization, facilitating embryo implantation, and maintaining fetal growth throughout pregnancy.

Anatomically, the system comprises paired **ovaries**, paired **fallopian tubes (oviducts)**, a central muscular **uterus**, a protective **cervix**, and the **vagina**. The uterine wall is composed of three distinct tissue layers: the inner vascularized **endometrium** (which undergoes monthly cyclic growth and shedding), the thick muscular **myometrium** (responsible for labor contractions), and the outer protective **perimetrium**.

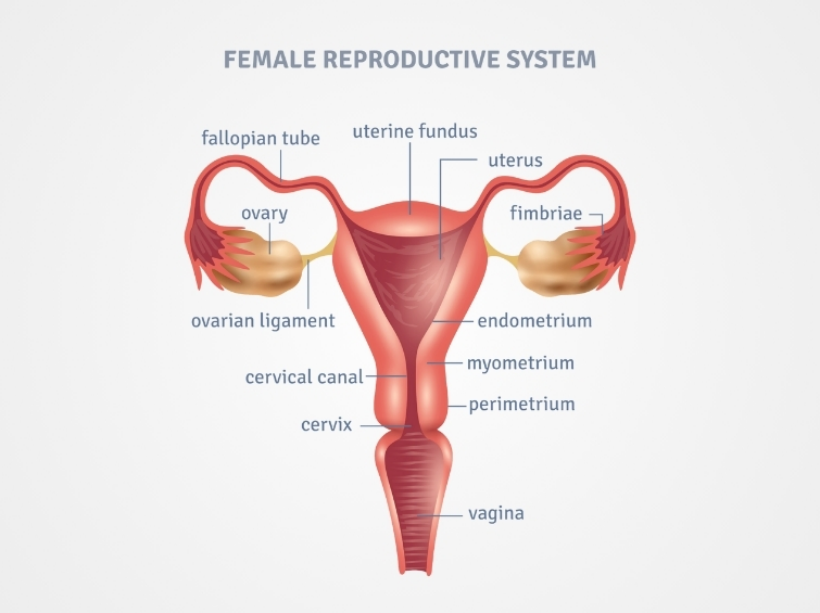


Figure 2: Female Reproductive System — Cross-Sectional Anatomical Diagram (FEMALE-REPRODUCTIVE-SYSTEM.jpg).

Anatomical Structure / Region	Physiological Function & Characteristics
UPPER STRUCTURES	
Ovary	Paired oval pelvic organs that produce egg cells (ova) and secrete estrogen and progesterone.
Ovarian ligament	Fibrous cord anchoring the ovary laterally to the uterine wall.
Fimbriae	Finger-like ciliated projections at the fallopian tube end that sweep released eggs into the lumen.
Fallopian tube	Muscular pathway transporting the ovum toward the uterus; primary site where fertilization occurs.
UTERUS & LAYERS	
Uterine fundus	The rounded superior region of the uterus located above the entry points of the fallopian tubes.
Uterus (Womb)	Hollow, pear-shaped muscular organ where a fertilized blastocyst implants and develops into a fetus.
Endometrium	Inner mucosal lining enriched with blood vessels; thickens monthly and sheds during menses.
Myometrium	Thick middle layer of smooth muscle tissue that generates powerful contractions during childbirth.
Perimetrium	Outer serous membrane covering the external surface of the uterus.
LOWER STRUCTURES	
Cervix	Narrow lower neck of the uterus projecting into the vagina; secretes protective mucus.
Cervical canal	Passageway connecting the uterine cavity with the vaginal lumen.
Vagina	Muscular canal serving as the female copulatory organ, birth canal, and menstrual exit path.

3. The Menstrual Cycle & Hormonal Feedback Loops

The menstrual cycle is a recurring ~28-day biological process in females that coordinates ovarian follicle development with uterine lining preparation under the strict control of the **Hypothalamic-Pituitary-Ovarian (HPO) axis**.

The endocrine cascade begins when the **hypothalamus** secretes pulsatile **Gonadotropin-Releasing Hormone (GnRH)**, stimulating the **anterior pituitary gland** to release **Follicle-Stimulating Hormone (FSH)** and **Luteinizing Hormone (LH)**. FSH drives primary ovarian follicles to mature and secrete rising levels of **estrogen**. High estrogen initially exerts negative feedback, but near Day 14, it triggers a dramatic **LH surge** that causes the dominant follicle to rupture, releasing the mature egg during **ovulation**. The ruptured follicle collapses into the **corpus luteum**, secreting high levels of **progesterone** to make the endometrium receptive. If fertilization fails, progesterone drops, triggering endometrial shedding.

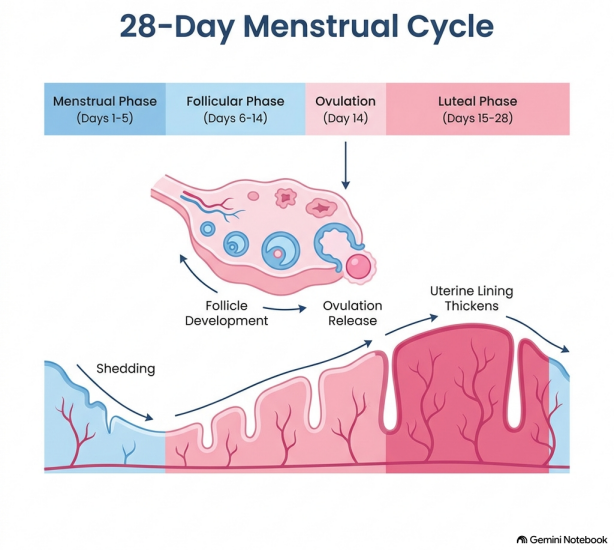


Figure 3: 28-Day Menstrual Cycle — Hormonal & Endometrial Profile Chart.

Cycle Phase	Timing	Ovarian Events	Uterine Events	Hormonal Dynamics
Menstrual Flow	Days 1–5	Early follicular recruitment under baseline FSH.	Degeneration and shedding of endometrial lining.	Estrogen & Progesterone drop to baseline levels.
Proliferative / Follicular	Days 6–13	Dominant follicle matures; secretes rising Estrogen.	Endometrium regrows, developing vascular supply.	Rising Estrogen causes positive feedback LH surge.
Ovulation Phase	Day 14	Mature follicle ruptures, releasing egg into tube.	Uterine lining continues thickening under Estrogen.	Sharp LH peak triggers follicular rupture.
Secretory / Luteal Phase	Days 15–28	Ruptured follicle converts to Corpus Luteum.	Endometrium becomes vascular, glandular, & receptive.	High Progesterone & Estrogen; drops if unfertilized.

4. Female Reproductive Health & Clinical Conditions

Pathological disruptions in the female reproductive tract encompass malignant growths, ectopic tissue proliferation, and endocrine cycle dysregulation. Understanding these conditions enables early clinical screening and effective therapeutic intervention.

Cervical Cancer Pathology: Cervical cancer arises from malignant cellular transformation within the cervical epithelium, primarily driven by high-risk Human Papillomavirus (HPV) infection, smoking, early sexual activity, and prolonged oral contraceptive use. Clinical signs include post-coital bleeding, irregular intermenstrual spotting, and foul-smelling vaginal discharge. Routine screening via **Pap smears** and **HPV DNA testing** detects pre-cancerous dysplasia early, allowing curative surgical excision or chemoradiation.

Endometriosis & Menstrual Abnormalities: Endometriosis is characterized by the ectopic growth of functional endometrial tissue outside the uterine cavity (e.g., on ovaries, fallopian tubes, and pelvic peritoneum). This tissue responds to cyclic ovarian hormones, undergoing monthly bleeding that causes severe pelvic inflammation, adhesions, chronic dysmenorrhea, and potential infertility.

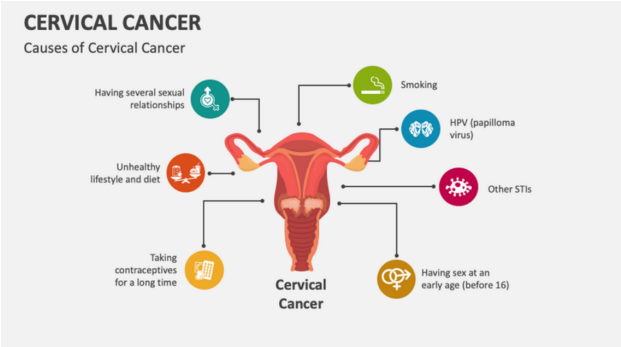


Figure 4.1: Cervical Cancer Primary Risk Factors, Diagnostic Screening & Treatment Pathways.

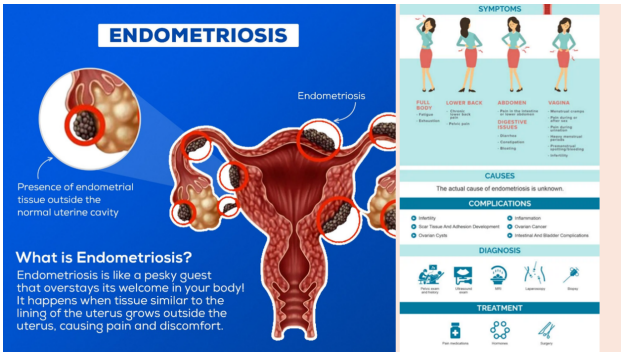


Figure 4.2: Endometriosis Pathophysiology & Ectopic Tissue Implants.

Condition Name	Clinical Features & Definition
Menorrhagia	Abnormally heavy or prolonged menstrual bleeding (>80 mL or >7 days duration).
Oligomenorrhea	Infrequent, delayed menstrual cycles occurring at intervals greater than 35 days.
Amenorrhea	Complete absence of menstruation (Primary: no menses by age 15; Secondary: absence >3 months).
Metrorrhagia	Irregular, unpredictable uterine bleeding occurring between expected menstrual periods.
Dysmenorrhea	Severe, painful uterine cramps during menses driven by excessive endometrial prostaglandins.
Polymenorrhea	Abnormally frequent menstrual cycles occurring at intervals less than 21 days.
Hypomenorrhea	Unusually light or scanty menstrual blood flow volume.
Adolescent Irregularity	Anovulatory, unpredictable cycles common during early post-menarche years.

5. Male Reproductive Health & Prostate Pathologies

Prostate gland disorders represent the most prevalent reproductive pathologies in aging men, ranging from malignant adenocarcinomas to acute or chronic glandular inflammation.

Prostate Cancer Pathology: Prostate cancer is a malignant adenocarcinoma developing in the peripheral zone of the prostate gland. As the tumor expands, it compresses the prostatic urethra, producing obstructive urinary symptoms including urinary hesitancy, weak stream, nocturia, dysuria, hematuria, and erectile dysfunction. Diagnosis involves Digital Rectal Examination (DRE), Prostate-Specific Antigen (PSA) blood testing, and transrectal ultrasound biopsy. Treatment options include radical prostatectomy, radiation, hormone deprivation therapy, and active surveillance.

Prostatitis Variations: Prostatitis involves inflammatory swelling of the prostate gland, categorized into four clinical types: **Acute Bacterial Prostatitis** (sudden fever, chills, severe dysuria), **Chronic Bacterial Prostatitis** (recurrent urinary tract infections), **Chronic Pelvic Pain Syndrome (CPPS)** (persistent pelvic/perineal pain without active bacterial infection), and **Asymptomatic Inflammatory Prostatitis** (white blood cells in semen without overt clinical pain).

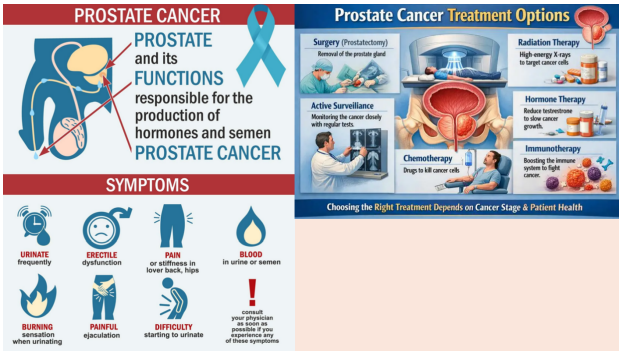


Figure 5.1: Prostate Cancer Pathology, Symptoms, & Preventive Lifestyle Strategies.

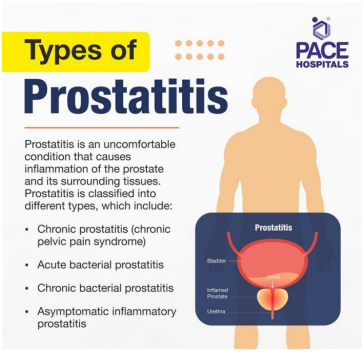


Figure 5.2: Prostatitis Anatomical Location & Clinical Category Classification.

Prostate Disorder	Clinical Category & Pathophysiology Summary
Prostate Cancer	Malignant prostate neoplasm; causes urethral compression, hematuria, and urinary retention.
Acute Bacterial Prostatitis	Sudden bacterial infection causing prostate swelling, high fever, chills, and severe dysuria.
Chronic Bacterial Prostatitis	Low-grade recurrent bacterial infection causing episodic pelvic pain and urinary distress.
Chronic Pelvic Pain Syndrome	Non-bacterial prostate pain syndrome causing chronic perineal, pelvic, and ejaculatory pain.
Asymptomatic Prostatitis	Inflammatory white blood cells present in prostatic secretions without overt pain symptoms.

6. The Complete Journey of Pregnancy: Conception to Embryogenesis

Human pregnancy begins with fertilizing a mature ovum and proceeds through a highly regulated sequence of cellular cleavage, blastocyst hatching, uterine implantation, gastrulation, and organogenesis.

Sperm Transport & Fertilization (Day 1): Following ejaculation, millions of sperm travel upward through the vagina, cervical canal, and uterine cavity into the fallopian tube. In the ampulla, capacitated sperm undergo the acrosome reaction. A single sperm penetrates the ovum's zona pellucida, fusing cell membranes and combining haploid genetic sets (23 paternal + 23 maternal) to form a single-celled diploid **Zygote**.

Cleavage, Morula & Blastocyst Stages (Days 2–5): As the zygote travels down the fallopian tube, it undergoes rapid mitotic cleavage (2, 4, 8, 16 cells). By Days 3–4, it forms a solid ball of cells called the **Morula**. By Day 5, fluid accumulates internally to form the **Blastocyst**, consisting of an **Inner Cell Mass (Embryoblast)** that becomes the embryo and an outer **Trophoblast** layer that forms the placenta.

Implantation & hCG Signalling (Days 6–12): On Days 6–7, the blastocyst adheres to and embeds into the vascular endometrial lining. Trophoblastic cells secrete **human Chorionic Gonadotropin (hCG)**, which signals the corpus luteum to continue progesterone production, maintaining the uterine lining and preventing menses.

Gastrulation & Germ Layers (Week 3): During Week 3, embryonic cells reorganize during **Gastrulation** to form three primary germ layers: **Ectoderm** (forms nervous system, brain, and skin), **Mesoderm** (forms heart, circulatory system, muscles, bones, and kidneys), and **Endoderm** (forms gastrointestinal tract lining, liver, pancreas, and lungs).

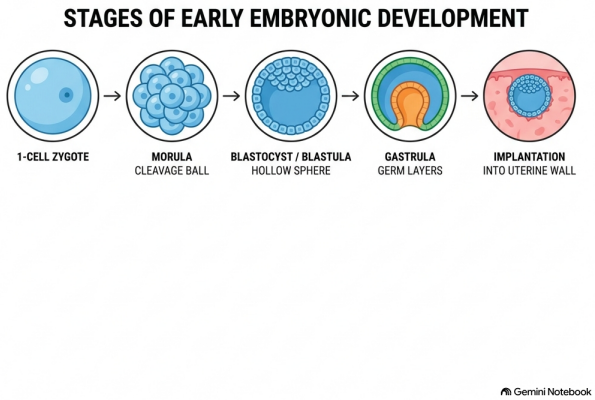


Figure 6.1: Step-by-Step Embryonic Development: Zygote to Gastrulation.

7. Fetal Maturation, Placental Support & Birth Milestones

Following embryogenesis, the developing organism enters the fetal period (Weeks 9–40), characterized by rapid tissue growth, functional organ maturation, and preparation for extrauterine life.

First Trimester Organogenesis (Weeks 4–12): Major organ systems establish rapidly. The neural tube closes, limb buds emerge, and the rudimentary tubular heart begins beating around **Days 21–22** post-conception. By Week 12, all organ systems are formed, facial features are defined, and the fetus exhibits passive movements.

Second Trimester Expansion (Weeks 13–27): Characterized by rapid body lengthening, bone ossification, and muscle refinement. Maternal perception of fetal movement (**quickening**) occurs around Weeks 16–20. The **placenta** fully assumes hormone production (estrogen and progesterone), supporting pregnancy independent of the corpus luteum.

Third Trimester Final Maturation (Weeks 28–40): Focuses on brain tissue growth, subcutaneous fat accumulation, and pulmonary maturation. Type II alveolar cells synthesize **surfactant** to prevent lung collapse at birth. Full-term development is achieved between Weeks 37 and 40, culminating in labor and delivery.

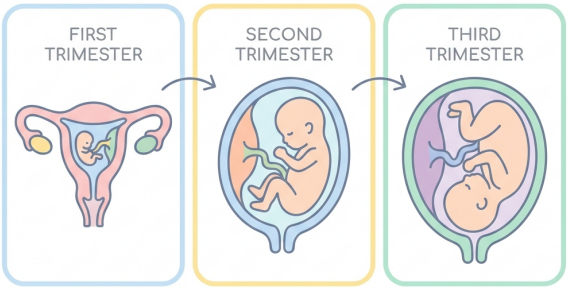


Figure 6.2: Fetal Growth & Trimester Development in the Womb.

Developmental Stage	Timeline	Key Biological Events & Morphological Milestones
Fertilization & Zygote	Day 1	Single sperm fuses with egg in fallopian ampulla; forms 1-cell Zygote (46 chromosomes).
Morula Stage	Days 3–4	Rapid mitotic cleavage produces a compact solid ball of 16–32 cells.
Blastocyst Stage	Day 5	Forms fluid-filled cavity, Inner Cell Mass (embryo), and outer Trophoblast (placenta).
Implantation	Days 6–7	Trophoblast embeds into endometrium; secretes hCG to maintain progesterone.
Gastrulation	Week 3	Inner cell mass differentiates into Ectoderm, Mesoderm, and Endoderm germ layers.
Organogenesis (1st Trim.)	Weeks 4–12	Heart beats at Day 21–22; neural tube closes; all major organ foundations form.
Fetal Growth (2nd Trim.)	Weeks 13–27	Rapid skeletal growth; bone ossification; quickening felt by mother; placenta fully active.
Final Maturation (3rd Trim.)	Weeks 28–40	Surfactant synthesis in lungs; rapid fat deposition and brain surface folding; full-term birth.